

Basis Sets & Integrals

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ESQC 2026



Expanding the molecular orbitals

Linear combination of basis functions

$$\psi_p(\mathbf{r}) = \sum_{\mu=1}^K C_{\mu p} \phi_{\mu}(\mathbf{r})$$

- The basis functions $\{\phi_{\mu}\}$ are **fixed**; the coefficients $C_{\mu p}$ are determined **variationally**
- Increasing K improves the flexibility and, ideally, approaches the **complete basis set limit (CBS)**
- Basis sets contain the **only empirical parameters** of an *ab initio* calculation

A useful basis set should be...

Accurate Flexible enough to describe the relevant occupied and virtual orbitals

Compact Able to reach the target accuracy with as few functions as possible

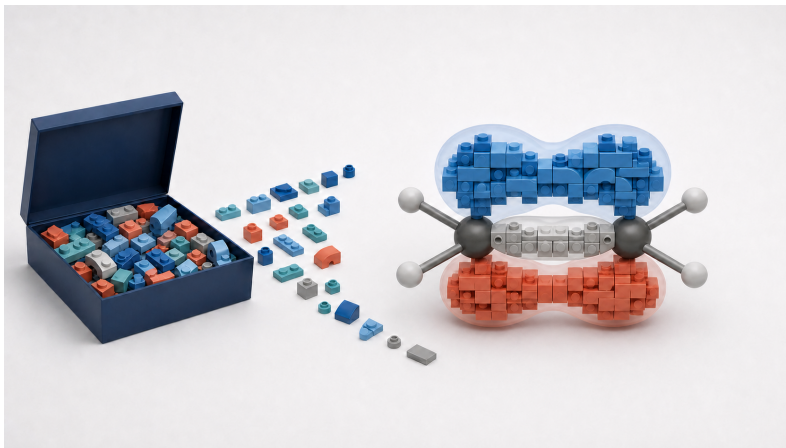
Stable Free from severe linear dependencies

Practical Compatible with evaluation of multicenter one- and two-electron integrals

⇒ **Basis set design is a compromise between accuracy and cost**

Basis functions are like LEGO bricks!

Key idea: Each basis function is a building block



Common basis functions

Gaussian-type orbitals (GTOs)

$$\phi(\mathbf{r}) = \exp(-\alpha r^2)$$

- Usually atom-centered
- Standard choice for molecular calculations
- Efficient evaluation of multicenter integrals
- Software: Orca, Q-Chem, Molcas, Molpro, PySCF, Psi4, CFOUR, MRCC, GAMESS, etc

Plane waves (PWs)

$$\phi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r})$$

- Delocalized and orthonormal
- Naturally adapted to periodic systems
- Systematically converged with an energy cutoff
- Software: Quantum ESPRESSO, VASP, ABINIT, CASTEP, etc

Slater-type orbitals (STOs)

$$\phi(\mathbf{r}) = \exp(-\zeta r)$$

- Usually atom-centered
- Natural nuclear cusp and exponential tail
- Software: ADF

Numerical atomic orbitals (NAOs)

- Radial functions tabulated numerically
- Compact and flexible atom-centered basis
- Software: FHI-aims, SIESTA, etc

Definition

$$\phi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \exp(i\mathbf{k} \cdot \mathbf{r}) \quad \text{with} \quad |\mathbf{k}| \leq k_{\text{cut}} \quad \text{and} \quad E_{\text{cut}} = \frac{1}{2} k_{\text{cut}}^2$$

Reciprocal-lattice vector

$$\mathbf{k} = k_1 \mathbf{b}_1 + k_2 \mathbf{b}_2 + k_3 \mathbf{b}_3$$

Unit cell

$$\underbrace{\Omega}_{\text{volume}} = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) \quad \text{with} \quad \mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$$

Vectors of the reciprocal lattice

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\Omega}$$

$$\mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\Omega}$$

$$\mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\Omega}$$

Integrals in the plane wave basis

Overlap integrals

$$\langle \mathbf{k}_1 | \mathbf{k}_2 \rangle = \int \phi_{\mathbf{k}_1}^*(\mathbf{r}) \phi_{\mathbf{k}_2}(\mathbf{r}) d\mathbf{r} = \frac{1}{\Omega} \int e^{i(\mathbf{k}_2 - \mathbf{k}_1) \cdot \mathbf{r}} d\mathbf{r} = \delta_{\mathbf{k}_1 \mathbf{k}_2} \Rightarrow \text{Orthogonal basis!}$$

Kinetic integrals

$$\langle \mathbf{k}_1 | -\frac{\nabla^2}{2} | \mathbf{k}_2 \rangle = \int \phi_{\mathbf{k}_1}^*(\mathbf{r}) \left(-\frac{\nabla^2}{2} \right) \phi_{\mathbf{k}_2}(\mathbf{r}) d\mathbf{r} = \int \phi_{\mathbf{k}_1}^*(\mathbf{r}) \frac{\mathbf{k}_2^2}{2} \phi_{\mathbf{k}_2}(\mathbf{r}) d\mathbf{r} = \frac{\mathbf{k}_2^2}{2} \delta_{\mathbf{k}_1 \mathbf{k}_2} \Rightarrow \text{Diagonal!}$$

Electron repulsion integrals (ERIs)

$$\begin{aligned} \langle \mathbf{k}_1 \mathbf{k}_2 | \mathbf{k}_3 \mathbf{k}_4 \rangle &= \iint \phi_{\mathbf{k}_1}^*(\mathbf{r}_1) \phi_{\mathbf{k}_2}^*(\mathbf{r}_2) r_{12}^{-1} \phi_{\mathbf{k}_3}(\mathbf{r}_1) \phi_{\mathbf{k}_4}(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \\ &= \frac{1}{\Omega^2} \iint \frac{e^{i(\mathbf{k}_3 - \mathbf{k}_1) \cdot \mathbf{r}_1} e^{i(\mathbf{k}_4 - \mathbf{k}_2) \cdot \mathbf{r}_2}}{r_{12}} d\mathbf{r}_1 d\mathbf{r}_2 \end{aligned}$$

Resolution of the Coulomb operator

Coulomb operator

$$\frac{1}{r_{12}} = \frac{1}{\Omega} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)}$$

Dealing with the points where $\mathbf{k} = 0$ is a bit tricky but doable...

Electron repulsion integrals

$$\begin{aligned} \langle \mathbf{k}_1 \mathbf{k}_2 | \mathbf{k}_3 \mathbf{k}_4 \rangle &= \frac{1}{\Omega^3} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} \iint e^{i(\mathbf{k}_3 - \mathbf{k}_1 - \mathbf{k}) \cdot \mathbf{r}_1} e^{i(\mathbf{k}_4 - \mathbf{k}_2 + \mathbf{k}) \cdot \mathbf{r}_2} d\mathbf{r}_1 d\mathbf{r}_2 \\ &= \frac{1}{\Omega} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} \delta_{\mathbf{k}_3 - \mathbf{k}_1 - \mathbf{k}, 0} \delta_{\mathbf{k}_4 - \mathbf{k}_2 + \mathbf{k}, 0} \\ &= \frac{4\pi}{\Omega |\mathbf{k}_1 - \mathbf{k}_3|^2} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} \Rightarrow \text{Momentum conservation!} \end{aligned}$$

Key idea: Assemblies of atoms are slight distortions of free electrons

Advantages

- ✓ Orthonormal and independent of atomic positions
- ✓ Naturally compatible with periodic boundary conditions
- ✓ Systematic convergence through a single parameter E_{cut}
- ✓ Kinetic and Coulomb operators are diagonal in reciprocal space
- ✓ Efficient evaluation using fast Fourier transforms
- ✓ No basis-set superposition error or atom-centered Pulay forces

Limitations

- ✗ Large basis sets:

$$N_{\text{PW}} \propto \Omega E_{\text{cut}}^{3/2}$$

- ✗ Core orbitals require prohibitively large cutoffs $\Rightarrow$ pseudopotentials
- ✗ Computational cost increases with supercell size
- ✗ Delocalized basis functions provide little real-space sparsity
- ✗ Large virtual spaces make correlated methods expensive

$\Rightarrow$ Particularly effective for periodic valence-electron calculations

Slater-type orbitals (STOs)

$$\phi_{\zeta n \ell m}^{\text{STO}}(\mathbf{r}) \propto r^{n-1} e^{-\zeta r} Y_{\ell m}(\hat{\mathbf{r}})$$

- ✓ It can model the electron-nucleus cusp!

$$\phi_{1s}^{\text{STO}}(r) \propto e^{-\zeta r} \Rightarrow \left. \frac{d\phi_{1s}^{\text{STO}}(r)}{dr} \right|_{r=0} = -\zeta$$

- ✓ Correct long-range behavior for IP!

$$\phi_{\text{HOMO}}^{\text{STO}}(r) \propto e^{-\sqrt{-2\epsilon_{\text{HOMO}}}r} \Rightarrow \text{IP} = -\epsilon_{\text{HOMO}}$$

- ✗ The multicenter integrals are **hard** to compute!

Gaussian-type orbitals (GTOs)

$$\phi_{\alpha n \ell m}^{\text{GTO}}(\mathbf{r}) \propto r^{2(n-1)-\ell} e^{-\alpha r^2} Y_{\ell m}(\hat{\mathbf{r}})$$

- ✗ They are flat at the electron-nucleus coalescence point!

$$\phi_{1s}^{\text{GTO}}(r) \propto e^{-\alpha r^2} \Rightarrow \left. \frac{d\phi_{1s}^{\text{GTO}}(r)}{dr} \right|_{r=0} = 0$$

- ✗ They decay too rapidly at large r !

- ✓ The multicenter integrals are **easy** to compute!

⇒ GTOs usually win but we use more!

Cartesian Gaussians

$$\text{Primitive GTO} = |\mathbf{a}\rangle = (x - A_x)^{a_x} (y - A_y)^{a_y} (z - A_z)^{a_z} e^{-\alpha |\mathbf{r} - \mathbf{A}|^2}$$

$$\text{Contracted GTO} = |\mathbf{a}\rangle \equiv \phi_{\mathbf{a}}^{\mathbf{A}}(\mathbf{r}) = \sum_k^K D_k |\mathbf{a}\rangle_k$$

- **Exponent:** α
 - **Center:** $\mathbf{A} = (A_x, A_y, A_z)$
 - **Angular momentum:** $\mathbf{a} = (a_x, a_y, a_z)$
 - **Total angular momentum:** $a = a_x + a_y + a_z$
-
- s shell: $s = (0, 0, 0)$
 - p shell: $p_x = (1, 0, 0)$ $p_y = (0, 1, 0)$ $p_z = (0, 0, 1)$
 - d shell: $(2, 0, 0), (0, 2, 0), (0, 0, 2), (1, 1, 0), (1, 0, 1), (0, 1, 1) \Rightarrow 6 \text{ } d \text{ cartesian functions!}$
 - f shell: $(3, 0, 0), (0, 3, 0), (0, 0, 3), (2, 1, 0), (2, 0, 1), (1, 2, 0), (0, 2, 1), (0, 1, 2), (1, 0, 2), (1, 1, 1) \Rightarrow 10 \text{ } f \text{ cartesian functions!}$

Cartesian vs Spherical Gaussians

Cartesian Gaussians

For a shell of angular momentum ℓ

$$N_{\ell}^{\text{cart}} = \frac{(\ell + 1)(\ell + 2)}{2}$$

ℓ	s	p	d	f
N_{ℓ}^{cart}	1	3	6	10

For instance, a Cartesian d shell contains

$$\{x^2, y^2, z^2, xy, xz, yz\}$$

Spherical Gaussians

For a pure angular momentum ℓ

$$N_{\ell}^{\text{sph}} = 2\ell + 1$$

ℓ	s	p	d	f
N_{ℓ}^{sph}	1	3	5	7

The extra Cartesian functions correspond to lower-angular-momentum components:

$$6d = 5d + 1s \quad 10f = 7f + 3p$$

$x^2 + y^2 + z^2 = r^2 \Rightarrow$ the Cartesian d shell contains an s -type component

$\Rightarrow$ Most quantum-chemistry calculations use spherical (pure) functions

The contraction problem

Primitive vs Contracted

- Same center A
- Same angular momentum α
- Different exponent α_k
- Contraction coefficient D_k and degree K

$$\underbrace{\langle \mathbf{ab} | \mathbf{cd} \rangle}_{\text{contracted ERI}} = \sum_i^{K_A} \sum_j^{K_B} \sum_k^{K_C} \sum_l^{K_D} D_i D_j D_k D_l \underbrace{[\mathbf{a}_i \mathbf{b}_j | \mathbf{c}_k \mathbf{d}_l]}_{\text{primitive ERI}}$$

One contracted ERI required $K_A \times K_B \times K_C \times K_D$ primitive ERIs!

$$|\mathbf{a}\rangle = \sum_k^K D_k |\mathbf{a}_k\rangle$$

<https://www.basissetexchange.org>

C	0	
S	10	1.00
	8.236000D+03	5.310000D-04
	1.235000D+03	4.108000D-03
	2.808000D+02	2.108700D-02
	7.927000D+01	8.185300D-02
	2.559000D+01	2.348170D-01
	8.997000D+00	4.344010D-01
	3.319000D+00	3.461290D-01
	9.059000D-01	3.937800D-02
	3.643000D-01	-8.983000D-03
	1.285000D-01	2.385000D-03

Dunning's cc-pVTZ basis for the carbon atom

$$\langle 1s1s | 1s1s \rangle = \sum_i^{10} \sum_j^{10} \sum_k^{10} \sum_l^{10} D_i D_j D_k D_l [s_i^{\alpha_i} s_j^{\alpha_j} | s_k^{\alpha_k} s_l^{\alpha_l}]$$

The $\langle 1s1s | 1s1s \rangle$ integral requires 10^4 s-type integrals!

Choose the basis for the physics you need

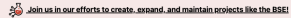
Basis set choice is tricky

- **Valence flexibility:** use double-, triple-, or higher- ζ descriptions to give each valence orbital several radial shapes
- **Polarization functions:** add **higher angular momenta** to describe anisotropic density, bonding, and angular correlation
- **Diffuse functions:** add **small exponents** to describe long-range density (anions, Rydberg states, noncovalent interactions, multipoles, (hyper)polarizabilities, etc)
- **Tight functions:** add **large exponents** to describe core region (core correlation, nuclear magnetic shielding, spin-spin coupling constants, etc)

⇒ The appropriate basis depends on both the electronic structure method and the property of interest!

Basis Set Exchange

 **Basis Set Exchange** [Download](#) [GitHub](#) [Feedback](#) [About](#) [Help](#)



All rolesAll

2ZaP
2ZaPa-NR
2ZaPa-NR-CV
3-21G
3ZaP
3ZaPa-NR
3ZaPa-NR-CV
4-31G
4ZaP
4ZaPa-NR
4ZaPa-NR-CV
5-21G
5ZaP
5ZaPa-NR
5ZaPa-NR-CV
6-21G
6-31+G
6-31+G*
6-31+G**
6-31+G**+J
6-31+G
6-31+G*
6-31+G**+J
6-31+G**

search basis sets...

References for selected basis

Plain TextGet References

Selected Basis Set:

Description:
Latest Version:
Last Update:
Role:
Family:
Function Types:

Basis Set NotesFamily Notes

Total found: 776 basis sets

Select AllReset Selection



In the selected basis set

Download basis set

Format: NWChemGet Basis SetAdvanced

Citation

When publishing results obtained from use of the Basis Set Exchange software, please cite:

- A New Basis Set Exchange: An Open, Up-to-date Resource for the Molecular Sciences Community. Benjamin P. Pritchard, Doaa Altarawy, Brett Didier, Tara D. Gibson, Theresa L. Windus. *J. Chem. Inf. Model.* **2019**, 59(7), 4814-4820, doi:10.1021/acs.jcim.9b00725.

For citing the previous EMSL/PHNL Basis Set Exchange, please cite the following references:

- The Role of Databases in Support of Computational Chemistry Calculations Feller, D., *J. Comp. Chem.* **1996**, 17(13), 1571-1586.
- Basis Set Exchange: A Community Database for Computational Sciences Schuchardt, K.L., Didier, B.T., Elsethagen, T., Sun, L., Gurumoorthy, V., Chase, J., Li, J., and Windus, T.L. *J. Chem. Inf. Model.* **2007**, 47(3), 1045-1052, doi:10.1021/ci600510i.

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<https://www.basissetexchange.org>

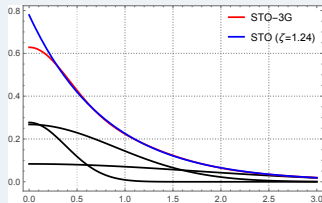
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Minimal basis set STO- n G

- One contracted basis function for each occupied orbital in the minimal atomic description
- Each contracted GTO is made of n primitive GTOs
- The contraction coefficients and exponents are chosen so that the contraction approximates a STO
- STO- n G basis sets are therefore **minimal**, **single- ζ** basis sets

STO-3G for the H atom

```
!-----  
! Basis Set Exchange  
! Version 0.12  
! https://www.basissetexchange.org  
!-----  
! Basis set: STO-3G  
! Description: STO-3G Minimal Basis (3 functions/AO)  
!           Role: orbital  
!           Version: 1 (Data from Gaussian09)  
!-----  
  
H      0  
S      3      1.00  
        0.3425250914D+01      0.1543289673D+00  
        0.6239137298D+00      0.5353281423D+00  
        0.1688554040D+00      0.4446345422D+00  
****
```



- 6-31+G*

- 6 six primitive GTOs for each core orbital
- 31 two sets of contracted GTOs (double- ζ) for each valence orbital with contraction degrees of 3 and 1
- + one set of diffuse functions on non-H atoms
- * one set of polarization functions on non-H atoms [$*$ $\equiv$ (d)]

- 6-311++G**

- 311 three sets of contracted GTOs with different exponents (triple- ζ)
- ++ same as + but with diffuse functions on H atoms
- ** same as * but with polarization functions on H atoms

- 6-311G(2df,2pd)

- 2df two sets of d 's and one set of f 's for non-H atoms
- 2pd two sets of p 's and one set of d 's for the H atoms

Correlation-consistent (cc) hierarchy

cc-pVXZ

$X = \text{D, T, Q, 5, 6, } \dots$

- Designed primarily for correlated wave function methods
- Polarization functions are added in balanced groups
- Increasing X systematically improves the valence space

Two-point extrapolation model

$$E_c(X) \approx E_c^{\text{CBS}} + AX^{-3}$$

Diffuse density

- aug-cc-pVXZ
- d-aug-cc-pVXZ
- t-aug-cc-pVXZ

Core correlation

- cc-pCVXZ: Optimized to recover both core-core and core-valence correlation
- cc-pwCVXZ: Uses a weighted optimization that emphasizes core-valence correlation

Correlation consistent basis for 2nd-row atoms

cc-pVDZ: 3s2p1d

s	s	s
p	p	
d		

aug-cc-pVDZ: 4s3p2d

s	s	s	s
p	p	p	
d	d		

d-aug-cc-pVDZ: 5s4p3d

s	s	s	s	s
p	p	p	p	
d	d	d		

cc-pVTZ: 4s3p2d1f

s	s	s	s
p	p	p	
d	d		
f			

aug-cc-pVTZ: 5s4p3d2f

s	s	s	s	s
p	p	p	p	
d	d	d		
f	f			

d-aug-cc-pVTZ: 6s5p4d3f

s	s	s	s	s	s
p	p	p	p	p	
d	d	d	d		
f	f	f			

cc-pVQZ: 5s4p3d2f1g

s	s	s	s	s
p	p	p	p	
d	d	d		
f	f			
g				

aug-cc-pVQZ: 6s5p4d3f2g

s	s	s	s	s	s
p	p	p	p	p	
d	d	d	d		
f	f	f			
g	g				

d-aug-cc-pVQZ: 7s6p5d4f3g

s	s	s	s	s	s	s
p	p	p	p	p	p	
d	d	d	d	d		
f	f	f	f			
g	g	g				

Core-valence correlation consistent basis for 2nd-row atoms

cc-pVDZ: $3s2p1d = 1s + 2s2p1d$

s	s	s
p	p	
d		

cc-pCVDZ: $4s3p1d = 2s1p + 2s2p1d$

s	s	s	s
p	p	p	
d			

cc-pVTZ: $4s3p2d1f = 1s + 3s3p2s1f$

s	s	s	s
p	p	p	
d	d		
f			

cc-pCVTZ: $6s5p3d1f = 3s2p1d + 3s3p2d1f$

s	s	s	s	s	s
p	p	p	p	p	
d	d	d			
f					

cc-pVQZ: $5s4p3d2f1g = 1s + 4s4p3d2f1g$

s	s	s	s	s
p	p	p	p	
d	d	d		
f	f			
g				

cc-pCVQZ: $8s7p5d3f1g = 4s3p2d1f + 4s4p3d2f1g$

s	s	s	s	s	s	s	s
p	p	p	p	p	p	p	
d	d	d	d	d			
f	f	f					
g							

Jensen basis sets

pc-0: 3s2p (unpolarized)

s s s
p p

pc-2: 4s3p2d1f

s s s s
p p p
d d
f

pc-1: 3s2p1d (one set of polarization)

s s s
p p
d

pc-3: 6s5p4d2f1g

s s s s s s
p p p p p
d d d d
f f
g

Polarization-consistent (pc) basis sets

- Designed for efficient convergence of HF and KS-DFT calculations
- Higher angular momenta are selected according to their energetic importance
- Diffuse variants: aug-pc-*n*

- Converge monotonically to the HF limit:

$$E_{\text{HF}}(L) = E_{\text{HF}}^{\text{CBS}} + A(L+1) \exp(-B\sqrt{L})$$

Gaussian basis sets: strengths and limitations

Key idea: Molecules are assemblies of slightly distorted atoms

Advantages

- ✓ Compact representation of molecular orbitals
- ✓ Efficient analytic evaluation of multicenter integrals
- ✓ Practical all-electron calculations, especially for light and medium elements
- ✓ Locality enables integral screening and sparse algorithms
- ✓ Well suited to mean-field and correlated methods

Limitations

- ✗ Nonorthogonal basis: possible linear-dependence problems
- ✗ Diffuse functions required for anions, Rydberg states, and long-range density tails
- ✗ Individual Gaussians have neither the correct nuclear cusp nor exponential asymptotic decay
- ✗ Atom-centered basis produces Pulay terms when nuclei move

⇒ Particularly effective for localized electronic structure correlated molecular calculations

Basis-set superposition error (BSSE)

Notation

We denote by

$$E_X^Y = \text{energy of system } X \text{ computed in the basis of system } Y$$

For example, E_A^{AB} is the energy of fragment A computed using the basis functions centered on both A and B . The basis functions on B are then **ghost functions** (no nucleus and no electrons on B).

$$E_A^A$$



basis on A only

$$E_A^{AB}$$



basis on A + **ghost basis** on B

$$E_{AB}^{AB}$$



full dimer basis

$$E_A^{AB} \leq E_A^A \Rightarrow \text{fragment } A \text{ is artificially stabilized by the basis on } B$$

$\Rightarrow$ Fragments can “borrow” basis functions from each other

Counterpoise correction

Boys-Bernardi counterpoise correction

With a finite atom-centered basis, each fragment can **borrow basis functions from the other fragment**

$$E_A^{AB} \leq E_A^A \quad \text{and} \quad E_B^{AB} \leq E_B^B$$

The usual interaction energy is

$$\Delta E_{\text{int}} = E_{AB}^{AB} - E_A^A - E_B^B$$

while the counterpoise-corrected interaction energy is

$$\Delta E_{\text{int}}^{\text{CP}} = E_{AB}^{AB} - E_A^{AB} - E_B^{AB}$$

Basis-set superposition error

$$\text{BSSE} = \Delta E_{\text{int}}^{\text{CP}} - \Delta E_{\text{int}} = \left(E_A^A - E_A^{AB} \right) + \left(E_B^B - E_B^{AB} \right) \geq 0$$

⇒ BSSE makes the uncorrected interaction energy artificially too attractive

Effective core potentials (ECPs) or pseudopotentials

Key idea: Treat only the chemically relevant electrons explicitly

One-electron ECP

Replace the core electrons by an **effective one-electron potential** acting on the remaining electrons

$$\hat{h}(\mathbf{r}) = -\frac{\nabla^2}{2} - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \quad \longrightarrow \quad \boxed{\hat{h}^{\text{ECP}}(\mathbf{r}) = -\frac{\nabla^2}{2} - \sum_A \frac{Q_A}{|\mathbf{r} - \mathbf{R}_A|} + \sum_A \hat{U}_A^{\text{ECP}}(\mathbf{r})}$$

Here, $Q_A = Z_A - N_A^{\text{core}}$ is an **effective nuclear charge** and N_A^{core} is the number of electrons replaced on atom A

Why use ECPs?

- ✓ **Lower cost:** fewer electrons and no basis functions needed to represent core orbitals
- ✓ **Relativistic effects:** can incorporate scalar-relativistic effects and, with a suitable operator, spin-orbit coupling
- ✓ Used in both molecular and periodic calculations, with Gaussian or plane-wave basis sets
- ✗ Introduce new **one-electron integrals** that can be tricky to compute
- ✗ Core properties are not described explicitly, and accuracy depends on the ECP quality

ECPs: projected and unprojected terms

A common semilocal form: scalar ECP

$$\hat{U}^{\text{ECP}}(\mathbf{r}) = U_L(r) + \sum_{\ell=0}^{L-1} \sum_{m=-\ell}^{\ell} |Y_{\ell m}\rangle U_{\ell}(r) \langle Y_{\ell m}|$$

- $U_L(r)$: short-range **local** potential $\Leftrightarrow$ **unprojected part**
- $\{|Y_{\ell m}\rangle U_{\ell}(r) \langle Y_{\ell m}|\}_{0 \leq \ell < L}$: angular-momentum-dependent **semi-local** potentials $\Leftrightarrow$ **projected part**

Gaussian expansion

$$U_{\ell}(r) = \sum_{k=1}^{K_{\ell}} D_{\ell k} r^{n_{\ell k}} \exp(-\eta_{\ell k} r^2) \quad \ell = 0, \dots, L$$

- $D_{\ell k}$ and $\eta_{\ell k} > 0$ are contraction coefficients and Gaussian exponents
- $n_{\ell k}$: radial powers, often -2 , -1 , or 0
- The angular projectors complicate the integrals, but they remain **one-electron integrals**

Choosing an ECP: core size and families

Small core vs large core: Ag ($Z = 47$)

Model	Core electrons	Explicit electrons	Explicit shells
Large core	36	11	$4d^{10} 5s^1$
Small core	28	19	$4s^2 4p^6 4d^{10} 5s^1$

A **small-core ECP** retains semicore shells explicitly: more flexibility, at a higher computational cost

Common ECP families used with Gaussian basis sets

- **LANL and SBK/SBKJC**: established families; LANL2DZ combines an ECP with a double- ζ valence basis
- **Stuttgart/Cologne**: energy-consistent pseudopotentials with different core sizes and dedicated basis sets <https://www.tc.uni-koeln.de/PP/clickpse.en.html>
- **def2-ECP**: paired with the def2 basis sets of Weigend and Ahlrichs
- **Burkatzki-Filippi-Dolg (BFD)**: developed for quantum Monte Carlo calculations
- **Correlation-consistent ECPs (ccECP)**: developed for high accuracy correlated many-body methods <https://pseudopotentiallibrary.org>

Most of them have dedicated basis sets!

Properties of Gaussian functions

Gaussian product rule: “The product of two gaussians is a gaussian”

$$G_{\alpha, \mathbf{A}}(\mathbf{r}) = \exp\left(-\alpha|\mathbf{r} - \mathbf{A}|^2\right) \quad \text{and} \quad G_{\beta, \mathbf{B}}(\mathbf{r}) = \exp\left(-\beta|\mathbf{r} - \mathbf{B}|^2\right) \quad \text{then}$$

$$\boxed{G_{\alpha, \mathbf{A}}(\mathbf{r})G_{\beta, \mathbf{B}}(\mathbf{r}) = K_{\zeta, \mathbf{P}}G_{\zeta, \mathbf{P}}(\mathbf{r})} \quad \text{with} \quad \zeta = \alpha + \beta \quad \text{and} \quad \mathbf{P} = \frac{\alpha\mathbf{A} + \beta\mathbf{B}}{\alpha + \beta}$$

$$K_{\zeta, \mathbf{P}} = \exp\left(-\frac{\alpha\beta}{\alpha + \beta}|\mathbf{A} - \mathbf{B}|^2\right) \Rightarrow \text{decay exponentially fast with } |\mathbf{A} - \mathbf{B}|$$

Overlap

$$\begin{aligned} \int G_{\alpha, \mathbf{A}}(\mathbf{r})G_{\beta, \mathbf{B}}(\mathbf{r}) \, d\mathbf{r} &= K_{\zeta, \mathbf{P}} \int G_{\zeta, \mathbf{P}}(\mathbf{r}) \, d\mathbf{r} \\ &= K_{\zeta, \mathbf{P}} \left(\int_{-\infty}^{+\infty} e^{-\zeta x^2} \, dx \right) \left(\int_{-\infty}^{+\infty} e^{-\zeta y^2} \, dy \right) \left(\int_{-\infty}^{+\infty} e^{-\zeta z^2} \, dz \right) \\ &= K_{\zeta, \mathbf{P}} \left(\frac{\pi}{\zeta} \right)^{3/2} \Rightarrow \text{cartesian factorization!} \end{aligned}$$

One-electron integrals over Gaussian functions

Kinetic energy

$$\begin{aligned} -\frac{1}{2}\nabla^2 G_{\beta,\mathbf{B}}(\mathbf{r}) &= [3\beta - 2\beta^2|\mathbf{r} - \mathbf{B}|^2]G_{\beta,\mathbf{B}}(\mathbf{r}) \\ -\frac{1}{2}\int G_{\alpha,\mathbf{A}}(\mathbf{r})\nabla^2 G_{\beta,\mathbf{B}}(\mathbf{r})\,d\mathbf{r} &= \int G_{\alpha,\mathbf{A}}(\mathbf{r})[3\beta - 2\beta^2|\mathbf{r} - \mathbf{B}|^2]G_{\beta,\mathbf{B}}(\mathbf{r})\,d\mathbf{r} \\ &= \frac{\alpha\beta}{\zeta}\left[3 - 2\frac{\alpha\beta}{\zeta}|\mathbf{A} - \mathbf{B}|^2\right]K_{\zeta,\mathbf{P}}\left(\frac{\pi}{\zeta}\right)^{3/2} \end{aligned}$$

Nuclear attraction

For a nucleus of charge Z_C located at $\mathbf{C}$

$$\begin{aligned} -Z_C \int G_{\alpha,\mathbf{A}}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{C}|} G_{\beta,\mathbf{B}}(\mathbf{r})\,d\mathbf{r} &= -Z_C K_{\zeta,\mathbf{P}} \int \frac{e^{-\zeta|\mathbf{r} - \mathbf{P}|^2}}{|\mathbf{r} - \mathbf{C}|}\,d\mathbf{r} \\ &= -\frac{2Z_C}{\sqrt{\pi}} K_{\zeta,\mathbf{P}} \int_0^\infty dt \int e^{-\zeta|\mathbf{r} - \mathbf{P}|^2} e^{-t^2|\mathbf{r} - \mathbf{C}|^2}\,d\mathbf{r} \\ &= -\frac{2\pi Z_C}{\zeta} K_{\zeta,\mathbf{P}} F_0\left(\zeta|\mathbf{P} - \mathbf{C}|^2\right) \end{aligned}$$

Four-center electron repulsion integrals

Gaussian product rule for ERIs

Formally, one has to compute $O(K^4)$ ERIs!

$$\begin{aligned}(\mathbf{ab}|\mathbf{cd}) &= \iint G_{\alpha,\mathbf{A}}(\mathbf{r}_1)G_{\beta,\mathbf{B}}(\mathbf{r}_1)\frac{1}{r_{12}}G_{\gamma,\mathbf{C}}(\mathbf{r}_2)G_{\delta,\mathbf{D}}(\mathbf{r}_2)\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2 \\ &= K_{\zeta,\mathbf{P}}K_{\eta,\mathbf{Q}}\iint G_{\zeta,\mathbf{P}}(\mathbf{r}_1)\frac{1}{r_{12}}G_{\eta,\mathbf{Q}}(\mathbf{r}_2)\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2\end{aligned}$$

The number of “significant” ERIs in a large system is $O(K^2)$!

Shell pairs and shell quartets

- $[\mathbf{ab}]$ = single primitive bra and $[\mathbf{ab}]$ = entire primitive shell pair
- $[\mathbf{ab}|\mathbf{cd}]$ = single primitive integral and $[\mathbf{ab}|\mathbf{cd}]$ = entire primitive shell quartet
- $(\mathbf{ab})$ = single contracted bra and $(\mathbf{ab})$ = entire contracted shell pair
- $(\mathbf{ab}|\mathbf{cd})$ = single contracted integral and $(\mathbf{ab}|\mathbf{cd})$ = entire contracted shell quartet

Definition

$$\begin{aligned} [ss|ss] &= K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \iint G_{\zeta, \mathbf{P}}(\mathbf{r}_1) \frac{1}{r_{12}} G_{\eta, \mathbf{Q}}(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \\ &= K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \iint e^{-\zeta|\mathbf{r}_1 - \mathbf{P}|^2} \frac{1}{r_{12}} e^{-\eta|\mathbf{r}_2 - \mathbf{Q}|^2} d\mathbf{r}_1 d\mathbf{r}_2 \end{aligned}$$

Laplace transform of the Coulomb operator

$$\frac{1}{r_{12}} = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2 r_{12}^2} dt = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2(x_1-x_2)^2} e^{-t^2(y_1-y_2)^2} e^{-t^2(z_1-z_2)^2} dt$$

$$\begin{aligned} [ss|ss] &= \frac{2}{\sqrt{\pi}} K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \int_0^\infty dt \left(\int_{-\infty}^\infty \int_{-\infty}^\infty e^{-\zeta(x_1 - \mathbf{P}_x)^2} e^{-t^2(x_1-x_2)^2} e^{-\eta(x_2 - \mathbf{Q}_x)^2} dx_1 dx_2 \right) \\ &\quad \times \left(\int_{-\infty}^\infty \int_{-\infty}^\infty e^{-\zeta(y_1 - \mathbf{P}_y)^2} e^{-t^2(y_1-y_2)^2} e^{-\eta(y_2 - \mathbf{Q}_y)^2} dy_1 dy_2 \right) \\ &\quad \times \left(\int_{-\infty}^\infty \int_{-\infty}^\infty e^{-\zeta(z_1 - \mathbf{P}_z)^2} e^{-t^2(z_1-z_2)^2} e^{-\eta(z_2 - \mathbf{Q}_z)^2} dz_1 dz_2 \right) \end{aligned}$$

Fundamental ssss integral

Defining $\vartheta = \zeta\eta/(\zeta + \eta)$, we have

$$\begin{aligned}
 [ss|ss] &= \frac{2}{\sqrt{\pi}} K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \int_0^\infty \left(\frac{\pi}{\sqrt{\zeta\eta}} \frac{e^{-\vartheta(\mathbf{P}_x - \mathbf{Q}_x)^2 \frac{t^2}{t^2 + \vartheta}}}{\sqrt{1 + \frac{t^2}{\vartheta}}} \right) \left(\frac{\pi}{\sqrt{\zeta\eta}} \frac{e^{-\vartheta(\mathbf{P}_y - \mathbf{Q}_y)^2 \frac{t^2}{t^2 + \vartheta}}}{\sqrt{1 + \frac{t^2}{\vartheta}}} \right) \left(\frac{\pi}{\sqrt{\zeta\eta}} \frac{e^{-\vartheta(\mathbf{P}_z - \mathbf{Q}_z)^2 \frac{t^2}{t^2 + \vartheta}}}{\sqrt{1 + \frac{t^2}{\vartheta}}} \right) dt \\
 &= \frac{2\pi^{5/2}}{(\zeta\eta)^{3/2}} K_{\zeta, \mathbf{P}} K_{\eta, \mathbf{Q}} \int_0^\infty \frac{e^{-\vartheta|\mathbf{P} - \mathbf{Q}|^2 \frac{t^2}{t^2 + \vartheta}}}{\left(1 + \frac{t^2}{\vartheta}\right)^{3/2}} dt
 \end{aligned}$$

$$\text{Setting } t^2 = \vartheta \frac{u^2}{1 - u^2} \quad \Rightarrow \quad [ss|ss] = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta, \mathbf{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta, \mathbf{Q}} \underbrace{\int_0^1 e^{-\vartheta|\mathbf{P} - \mathbf{Q}|^2 u^2} du}_{F_0(\vartheta|\mathbf{P} - \mathbf{Q}|^2) = \text{Boys function}}$$

$$[00|00]^{(0)} = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta, \mathbf{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta, \mathbf{Q}} F_0(\vartheta|\mathbf{P} - \mathbf{Q}|^2)$$

Boys function

Generalized Boys function

$$F_m(x) = \int_0^1 t^{2m} \exp(-xt^2) dt \quad F_0(x) = \sqrt{\frac{\pi}{4x}} \operatorname{erf}(\sqrt{x})$$

Computation

- $F_m(x)$ computed using pretabulated values on a grid and Taylor expansion (Chebyshev interpolation table)
- lower m values computed via downward recursion

Downward recursion

$$F_m(x) = \frac{2xF_{m+1}(x) + \exp(-x)}{2m+1}$$

Downward recursion is numerically stable, whereas upward recursion can become unstable for small x

Asymptotic behavior

$$F_m(x) \sim \frac{(2m-1)!!}{2^{m+1}} \sqrt{\frac{\pi}{x^{2m+1}}} \quad \text{for large } x$$

Why do higher-order Boys functions appear?

The physical ssss ERI involves only F_0 .

However, generating angular momentum by differentiation produces higher-order Boys functions F_m . It is therefore convenient to define the hierarchy

$$[00|00]^{(m)} = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta, \mathbf{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta, \mathbf{Q}} F_m(\vartheta|\mathbf{P} - \mathbf{Q}|^2)$$

- $m = 0$: physical ssss ERI
- $m > 0$: auxiliary integrals, used only as intermediates
- Angular-momentum recurrences couple auxiliary orders m and $m + 1$
- For total angular momentum $L = a + b + c + d$
 $\Rightarrow$ one needs fundamental auxiliary integrals up to $m = L$

Key identity

For a Cartesian Gaussian,

$$\frac{\partial |\mathbf{a}|}{\partial A_i} = 2\alpha |\mathbf{a} + \mathbf{1}_i| - a_i |\mathbf{a} - \mathbf{1}_i| \quad \Rightarrow \quad |\mathbf{a} + \mathbf{1}_i| = \frac{1}{2\alpha} \left[\frac{\partial |\mathbf{a}|}{\partial A_i} + a_i |\mathbf{a} - \mathbf{1}_i| \right]$$

Applying this identity to the auxiliary ERIs generates recurrence relations connecting different angular momenta and different Boys orders $\Rightarrow$ **Obara-Saika or vertical recurrence relations**

$$[00|00]^{(m)} \longrightarrow [a0|00]^{(m)} \longrightarrow [a0|c0]^{(m)}$$

Instead of evaluating every ERI from scratch, we build them recursively

Strategy for building an ERI

Starting from the fundamental auxiliary integrals,

$$\boxed{[00|00]^{(m)} \xrightarrow{\text{VRR}} [a0|c0]^{(m)} \xrightarrow{\text{contraction}} (a0|c0) \xrightarrow{\text{HRR}} (ab|cd)}$$

- **Vertical recurrence relations (VRRs)**: create angular momentum on one center of each Gaussian pair
- **Contraction**: combine primitive integrals into contracted integrals
- **Horizontal recurrence relations (HRRs)**: redistribute angular momentum within each Gaussian pair

VRRs create angular momentum	HRRs redistribute it
------------------------------	----------------------

VRR on the bra pair: build angular momentum on A

$$\begin{aligned}
 [(\mathbf{a} + \mathbf{1}_i)\mathbf{0}|\mathbf{00}]^{(m)} &= (P_i - A_i)[\mathbf{a0}|\mathbf{00}]^{(m)} \\
 &\quad + \frac{\eta}{\zeta + \eta} (P_i - Q_i)[\mathbf{a0}|\mathbf{00}]^{(m+1)} \\
 &\quad + \frac{a_i}{2\zeta} \left[[(\mathbf{a} - \mathbf{1}_i)\mathbf{0}|\mathbf{00}]^{(m)} - \frac{\eta}{\zeta + \eta} [(\mathbf{a} - \mathbf{1}_i)\mathbf{0}|\mathbf{00}]^{(m+1)} \right]
 \end{aligned}$$

VRR on the ket pair: build angular momentum on C

$$\begin{aligned}
 [\mathbf{a0} | (\mathbf{c} + \mathbf{1}_i)\mathbf{0}]^{(m)} &= (Q_i - C_i)[\mathbf{a0}|\mathbf{c0}]^{(m)} \\
 &\quad + \frac{\zeta}{\zeta + \eta} (Q_i - P_i)[\mathbf{a0}|\mathbf{c0}]^{(m+1)} \\
 &\quad + \frac{c_i}{2\eta} \left[[\mathbf{a0} | (\mathbf{c} - \mathbf{1}_i)\mathbf{0}]^{(m)} - \frac{\zeta}{\zeta + \eta} [\mathbf{a0} | (\mathbf{c} - \mathbf{1}_i)\mathbf{0}]^{(m+1)} \right] \\
 &\quad + \frac{a_i}{2(\zeta + \eta)} [(\mathbf{a} - \mathbf{1}_i)\mathbf{0}|\mathbf{c0}]^{(m+1)}
 \end{aligned}$$

Horizontal recurrence relations

Redistributing angular momentum

Once angular momentum has been generated on one center of a Gaussian pair, it can be shifted to the other center:

$$[a(b + 1_i)|cd] = [(a + 1_i)b|cd] + (A_i - B_i)[ab|cd]$$

and, similarly, angular momentum can be transferred from C to D

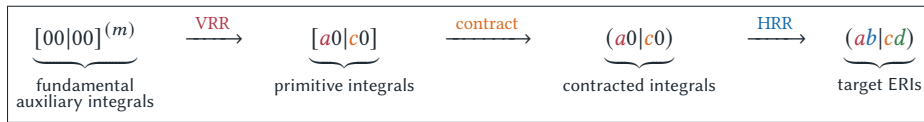
Why apply HRRs after contraction?

The HRR depends only on the geometry through $A_i - B_i$: it is independent of the Gaussian exponents

Therefore, the same recurrence applies directly to contracted integrals

$$(a(b + 1_i)|cd) = ((a + 1_i)b|cd) + (A_i - B_i)(ab|cd)$$

Late-contraction ERI algorithm



1. Compute the required Boys functions $F_m(T)$
2. Form the fundamental auxiliary integrals $[00|00]^{(m)}$
3. Use **VRRs** to build angular momentum on A and C
4. **Contract** the primitive integrals
5. Use exponent-independent **HRRs** to transfer angular momentum $A \rightarrow B$ and $C \rightarrow D$

Expensive exponent-dependent recurrences are performed before contraction, while the cheap exponent-independent HRRs are performed afterwards

A “good” upper bound must be

- simple, i.e, cheap to compute
- strong, i.e., a good estimate
- scaling-consistent, i.e., $N_{\text{sig}} = O(N_{\text{upper}})$

Cauchy-Schwartz bound

$$|(ab|cd)| \leq \sqrt{(ab|ab)}\sqrt{(cd|cd)} \quad \text{or} \quad |(P|Q)| \leq \sqrt{(P|P)}\sqrt{(Q|Q)}$$

- $(P|Q)$ should decay as $1/|P - Q|$ or faster
- However, there is no distance information in the product $(P|P)(Q|Q)$
- The Cauchy-Schwarz bound is rigorous and cheap but is not tight for well-separated shell pairs

⇒ A cheap upper bound avoids the evaluation of negligible ERIs

Integral bound

We seek a single number that bounds a particular integral

$$|(\textcolor{blue}{a}\textcolor{red}{b}|\textcolor{blue}{c}\textcolor{red}{d})| \leq \sqrt{(\textcolor{blue}{a}\textcolor{red}{b}|\textcolor{blue}{a}\textcolor{red}{b})} \sqrt{(\textcolor{blue}{c}\textcolor{red}{d}|\textcolor{blue}{c}\textcolor{red}{d})}$$

With $O(K^2)$ $(\textcolor{blue}{a}\textcolor{red}{b}|\textcolor{blue}{a}\textcolor{red}{b})$ bound factors, we can upper bound $O(K^4)$ $(\textcolor{blue}{a}\textcolor{red}{b}|\textcolor{blue}{c}\textcolor{red}{d})$ integrals

Class bound

We seek a single number that bounds an entire class

$$|(\textcolor{blue}{a}\textcolor{red}{b}|\textcolor{blue}{c}\textcolor{red}{d})| \leq Q_{\textcolor{blue}{a}\textcolor{red}{b}} Q_{\textcolor{blue}{c}\textcolor{red}{d}} \quad \text{with} \quad Q_{\textcolor{blue}{a}\textcolor{red}{b}} = \max_{\substack{\textcolor{red}{a} \in a \\ \textcolor{blue}{b} \in b}} \sqrt{(\textcolor{blue}{a}\textcolor{red}{b}|\textcolor{blue}{a}\textcolor{red}{b})}$$

Therefore,

if $Q_{\textcolor{blue}{a}\textcolor{red}{b}} Q_{\textcolor{blue}{c}\textcolor{red}{d}} < \tau$ then **discard the entire shell quartet!**

*The simple $(dd|dd)$ class is made of 1296 integrals!

Asymptotic scaling of two-electron integrals

Number of significant two-electron integrals

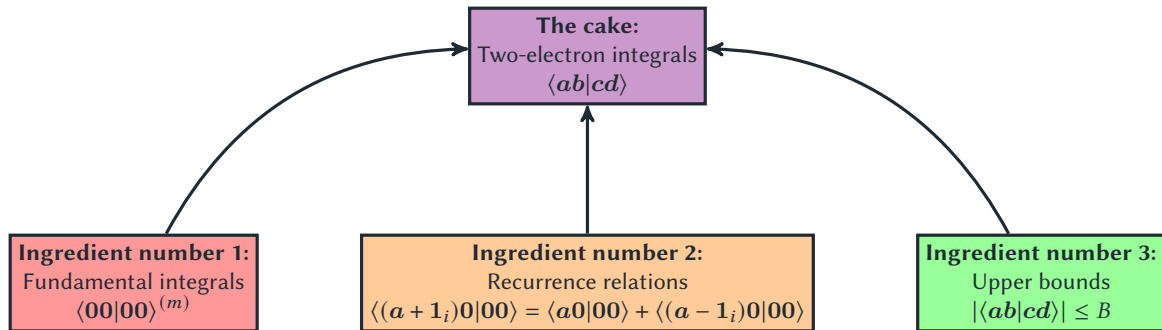
$$(ab|cd) \equiv (ab|\hat{O}_2|cd)$$

Long-range vs short-range operators

$$N_{\text{sig}} = c N^\alpha$$

Molecule	N	$\hat{O} = r_{12}^{-1}$		$\hat{O} = e^{-r_{12}^2}$	
		N_{sig}	α	N_{sig}	α
propene	12	1 625	—	1 650	—
butadiene	16	5 020	3.9	5 020	3.9
hexatriene	24	24 034	3.9	23 670	3.8
octatetraene	32	63 818	3.4	52 808	2.8
decapentaene	40	119 948	2.8	81 404	1.9
dodecaexaene	48	192 059	2.6	109 965	1.6

Recipe for computing two-electron integrals



Direct SCF algorithm

From ERIs to the Fock matrix

The Coulomb and exchange matrices are

$$J_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma)$$

$$K_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\lambda|\nu\sigma)$$

Direct SCF: trading disk storage for repeated integral evaluation

Instead of storing all $O(K^4)$ ERIs

1. loop over shell quartets $(\mu\nu|\lambda\sigma)$
2. use an upper bound to discard negligible quartets
3. compute the significant ERIs
4. immediately accumulate their contributions to $\mathbf{J}$ and $\mathbf{K}$
5. discard the ERIs

The density matrix elements can be included in the screening process

screen $\longrightarrow$ compute $\longrightarrow$ accumulate $\longrightarrow$ discard

Resolution of the identity (RI)

The product space is the problem

An ERI is the Coulomb interaction between two **pair densities**,

$$(\mu\nu|\lambda\sigma) = \iint \underbrace{\phi_\mu(\mathbf{r}_1)\phi_\nu(\mathbf{r}_1)}_{\rho_{\mu\nu}(\mathbf{r}_1)} \frac{1}{r_{12}} \underbrace{\phi_\lambda(\mathbf{r}_2)\phi_\sigma(\mathbf{r}_2)}_{\rho_{\lambda\sigma}(\mathbf{r}_2)} d\mathbf{r}_1 d\mathbf{r}_2$$

With K basis functions, there are formally $O(K^2)$ different pair densities $\rho_{\mu\nu}$

Density fitting

Instead of representing each product $\phi_\mu\phi_\nu$ exactly, we expand it in a smaller **auxiliary basis** $\{\varphi_A^{\text{aux}}\}$,

$$\rho_{\mu\nu}(\mathbf{r}) \approx \tilde{\rho}_{\mu\nu}(\mathbf{r}) = \sum_A C_{\mu\nu}^A \varphi_A^{\text{aux}}(\mathbf{r})$$

with typically $K_{\text{aux}} = O(K)$

$\Rightarrow$ RI replaces the large AO-product space by a compact auxiliary space

How should the pair density be fitted?

Define the Coulomb inner product

$$(f|g) = \iint f(\mathbf{r}_1) \frac{1}{r_{12}} g(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2$$

The optimal coefficients minimize the Coulomb norm of the fitting error,

$$\Delta_{\mu\nu} = (\rho_{\mu\nu} - \tilde{\rho}_{\mu\nu} | \rho_{\mu\nu} - \tilde{\rho}_{\mu\nu})$$

Least-squares equations

Minimization with respect to $C_{\mu\nu}^A$ gives

$$\sum_B \underbrace{(A|B)}_{V_{AB}} C_{\mu\nu}^B = \underbrace{(A|\mu\nu)}_{\text{three-center integral}} \Rightarrow \boxed{C_{\mu\nu}^A = \sum_B \left(\mathbf{V}^{-1} \right)_{AB} (B|\mu\nu)}$$

$\Rightarrow$ The Coulomb metric determines the optimal auxiliary expansion

RI approximation to the ERIs

From four-center to three-center integrals

Inserting the fitted pair densities gives

$$(\mu\nu|\lambda\sigma) \approx \sum_{AB} (\mu\nu|A) \left(\mathbf{V}^{-1} \right)_{AB} (B|\lambda\sigma)$$

Symmetric factorization

Defining the orthonormalized three-index quantities

$$B_{\mu\nu}^P = \sum_A (\mu\nu|A) \left(\mathbf{V}^{-1/2} \right)_{AP}$$

one obtains the particularly simple form

$$(\mu\nu|\lambda\sigma) \approx \sum_P B_{\mu\nu}^P B_{\lambda\sigma}^P$$

⇒ A four-index tensor is factorized into products of three-index tensors

Low-rank representations of the ERIs

Resolution of the identity = Density fitting

In RI/DF, the AO pair densities are expanded in a **predefined auxiliary basis** $\{\chi_A^{\text{aux}}\}$,

$$\chi_\mu(\mathbf{r})\chi_\nu(\mathbf{r}) \approx \sum_A C_{\mu\nu}^A \chi_A^{\text{aux}}(\mathbf{r}) \quad \Rightarrow \quad (\mu\nu|\lambda\sigma) \approx \sum_P B_{\mu\nu}^P B_{\lambda\sigma}^P$$

$\Rightarrow$ The quality of the approximation depends on the auxiliary basis

Cholesky decomposition (CD)

The ERI tensor $\mathbf{V}_{(\mu\nu),(\lambda\sigma)} = (\mu\nu|\lambda\sigma)$ can be viewed as a positive semidefinite matrix in the compound indices $(\mu\nu)$ and $(\lambda\sigma)$. A pivoted Cholesky decomposition directly generates

$$(\mu\nu|\lambda\sigma) \approx \sum_{P=1}^{N_{\text{CD}}} L_{\mu\nu}^P L_{\lambda\sigma}^P$$

until the remaining error is smaller than a prescribed threshold τ_{CD}

$\Rightarrow$ RI/DF uses a predefined auxiliary space; CD constructs one adaptively

Construction of the Coulomb matrix

Starting from

$$J_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma),$$

the RI approximation gives

$$J_{\mu\nu} \approx \sum_{AB} (\mu\nu|A) \underbrace{\left(\mathbf{V}^{-1} \right)_{AB} \sum_{\lambda\sigma} P_{\lambda\sigma} (B|\lambda\sigma)}_{d_B}.$$

Thus,

$$\boxed{\mathbf{P} \longrightarrow \{d_A\} \longrightarrow \{c_A\} = \mathbf{V}^{-1}\{d_A\} \longrightarrow \mathbf{J}}$$

- No need to store the full $\mathcal{O}(K^4)$ ERI tensor
- Only **two-center** ($A|B$) and **three-center** ($\mu\nu|A$) integrals are required
- Auxiliary basis sets are optimized to make the fitting error very small
- RI is also commonly called **density fitting (DF)**
 $\Rightarrow$ RI dramatically reduces the cost and memory requirements of Coulomb contractions